

Literature Already Answered: Set-Theoretic Dichotomy for TC/KR Density

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Status

This is a **literature-already-answered packet** for Remark 1.14 of Ostrovska and Ostrovskii, *Generalized transportation cost spaces*, arXiv:1902.10334 [1]. Raad’s characterization gives the exact criterion needed for the source question. The supporting paper does not appear to cite this source remark directly, so the identification below records the bridge: the completion of $\text{TP}(M)$ is the Lipschitz-free space $\mathcal{F}(M)$.

Source Question

The source paper proves Theorem 1.13: if M is Polish, then $(\text{TP}(M), \|\cdot\|_{\text{TC}})$ is dense in the Kantorovich–Rubinstein measure space $(\mathcal{M}, \|\cdot\|_{\text{KR}})$. Remark 1.14 says that the authors do not know how to prove an analogue for general metric spaces. In the local parsed source this is `data/parsed/arxiv_sources/1902.10334/source.tex`, lines 680–746, and in the PDF it is page 11.

Next, we split K into a finite number $\{K_n\}_{n \in T}$ of pairwise disjoint Borel subsets of diameter $< \varepsilon/(3(\mu(M) + \nu(M)))$ each. Define a function f on M as follows: In each of the sets K_n , we pick a point t_n and let $f(t_n) = \mu(K_n) - \nu(K_n)$ for $n \in T$. If f is extended as 0 to the rest of M , the obtained function is not necessarily in $\text{TP}(M)$. To balance this, the 0-extension is modified at point O by taking $f(O) = -\sum_{n \in T} f(t_n)$, thus implying $f \in \text{TP}(M)$. If $t_n \neq O$, the following inequality holds:

$$\begin{aligned} \left| \int_{K_n} l(v) d(\mu - \nu)(v) - l(t_n) f(t_n) \right| &= \left| \int_{K_n} (l(v) - l(t_n)) d(\mu - \nu)(v) \right| \\ &\leq \int_{K_n} |l(v) - l(t_n)| d\mu(v) + \int_{K_n} |l(v) - l(t_n)| d\nu(v) \leq \frac{\varepsilon}{3} \cdot \frac{\mu(K_n) + \nu(K_n)}{\mu(K) + \nu(K)}, \end{aligned}$$

where, in the last inequality, the fact that l is 1-Lipschitz and the assumption on the diameter of K_n are used.

Since $l(O) = 0$, one has:

$$\begin{aligned} |l(\mu - \nu) - l(f)| &\leq \sum_{n \in T} \left| \int_{K_n} l(v) d(\mu - \nu)(v) - l(t_n) f(t_n) \right| + \left| \int_{B(O, R) \setminus K} l(v) d\mu(v) \right| \\ &\quad + \left| \int_{B(O, R) \setminus K} l(v) d\nu(v) \right| + \left| \int_{M \setminus B(O, R)} l(v) d\mu(v) \right| + \left| \int_{M \setminus B(O, R)} l(v) d\nu(v) \right| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{6} + \frac{\varepsilon}{6} + \frac{\varepsilon}{6} + \frac{\varepsilon}{6} = \varepsilon. \end{aligned}$$

□

Remark 1.14. We do not know how to prove an analogue of Theorem [1.13](#) in the case of general metric spaces. See [\[Bil68\]](#) p. 234–235 in this connection. Nonseparability will not be an obstacle if we consider uniformly discrete spaces, because in such spaces finite Borel measures have countable support. The general metric case is beyond the subject of this work. For a systematic exposition of generalizations of results of Kantorovich and Rubinstein we refer the reader to [\[RR98\]](#).

Finally, it is worth mentioning that in Computer Science the transportation cost is often called *Earth Mover's Distance*. This name was introduced by Rubner-Tomasi-Guibas [\[RTG98\]](#) in their work on computer vision. They knew the notion of transportation cost and their goal was to generalize it to the cases where total demand can be less than the total supply. This more general case is not considered in the present paper as such generalized transportation problems do not form a vector space.

Several authors published their opinions on the most suitable choice of the name for the spaces which we call *transportation cost spaces*. Vershik [\[Ver04\]](#), [\[Ver13\]](#) provided an argument in favor of the name *Kantorovich space*. Villani [\[Vil09\]](#) pp. 106–107] decided in favor of Wasserstein, and Weaver [\[Wea18\]](#) p. 125] defended the name *Arens-Eells space*. It is interesting to mention that Villani decided in favor of

Supporting Theorem

The decisive supporting paper is Lucas Maciel Raad, *A Characterization of Borel Measures which Induce Lipschitz-Free Space Elements*, arXiv:2412.13319 [2]. Its Theorem 3.6 states that for a pointed metric space M and a Borel signed measure μ , the integration functional

$$L_\mu(f) = \int_M f d\mu, \quad f \in \text{Lip}_0(M),$$

belongs to $\mathcal{F}(M)$ if and only if

$$\int_M d(x, 0) d|\mu|(x) < \infty$$

and μ is concentrated on a separable subset of M . Its Corollary 3.7, using Theorem 2.16, says equivalently that every finite-first-moment Borel measure on M induces an element of $\mathcal{F}(M)$ if and only if there is no nontrivial measure on the weight of M .

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We can now show an example⁴ of a pointed metric space M and an element of $\mathcal{F}(M)$ that is induced by a Borel measure on $\overline{M}$ but is not induced by a Borel measure on M :

Example 3.5. Let $M = \{0\} \cup (1, 2]$. By a slight abuse of notation we have that $\mathcal{L}\delta_1 = \delta(1) \in \mathcal{F}(M)$. But if μ is a Borel measure on M such that $\mathcal{L}\mu \in \mathcal{F}(M)$ then $\mathcal{L}\mu \neq \delta(1)$.

Proof. Suppose by contradiction that μ is a Borel measure on M such that $\mathcal{L}(\mu) = \delta(1)$.

If we define μ_0 by $d\mu_0 = \chi_{M \setminus \{0\}} d\mu$ then $\mathcal{L}\mu = \mathcal{L}\mu_0$. Therefore we may assume without loss of generality that μ is concentrated on $(1, 2]$. First notice that μ is finite by Theorem 3.4(a). Now fix a closed $F \subset (1, 2]$ and denote $k = d(1, F)^{-1}$. We can define $f_n \in \text{Lip}_0(M)$ by:

$$f_n(m) = \max\left(0, (1 - k2^n d(m, F))\right)$$

for all $m \in M$. Therefore, by the Lebesgue dominated convergence theorem and the Radon-Nikodym theorem, we have that:

$$\mu(F) = \int \chi_F d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu = 0$$

But, μ is finite and therefore regular (see for example [5, Theorem 7.1.7]). From which we conclude that $\mu = 0$, a contradiction. $\square$

We can now show a sufficient and necessary condition for a Borel measure to induce an element of $\mathcal{F}(M)$:

Theorem 3.6. *Let M be a pointed metric space and let μ be a Borel measure on M . Then $\mathcal{L}\mu \in \mathcal{F}(M)$ if and only if $\int \rho d|\mu| < \infty$ and μ is concentrated on a separable subset of M .*

Proof. By Theorem 3.3 we only need to show that if M is a pointed metric space and μ is a Borel measure on M such that $\int \rho d|\mu| < \infty$, then $\mathcal{L}\mu \in \mathcal{F}(M)$ if and only if μ is concentrated on a separable subset of M .

$\Rightarrow$) Notice that if we define μ_0 by $d\mu_0 = \chi_{M \setminus \{0\}} d\mu$ then $\mathcal{L}\mu = \mathcal{L}\mu_0$.

Therefore we may assume without loss of generality that $\mu(\{0\}) = 0$.

By the hypothesis that $\mathcal{L}\mu \in \mathcal{F}(M)$, we can write $\mathcal{L}\mu$ in the form:

$$\mathcal{L}\mu = \lim_{n_1 \rightarrow \infty} \sum_{n_2 \in \mathbb{N}} a_{(n_1, n_2)} \delta(m_{(n_1, n_2)}),$$

where $a_n \in \mathbb{R}$ and $m_n \in M$ for all $n \in \mathbb{N}^2$ and for each $n_1 \in \mathbb{N}$ the set of the $n_2 \in \mathbb{N}$ for which $a_{(n_1, n_2)} \neq 0$ is finite.

⁴This is also basically the standard example of an incomplete metric space M such that there is a nonzero $m \in \mathcal{F}(M)$ having empty support. See [1, Proposition 3.3.10].

Identification

The completion of $(\text{TP}(M), \|\cdot\|_{\text{TC}})$ is the Lipschitz-free space $\mathcal{F}(M)$. A finite signed Borel measure σ of total mass zero and finite first moment determines an element of $\text{Lip}_0(M)^*$ by

$$f \mapsto \int_M f d\sigma.$$

Thus σ is approximable by finitely supported transportation problems in the free-space norm exactly when this functional belongs to $\mathcal{F}(M)$.

For the positive KR-norm statement, suppose every finite Borel measure on M is concentrated on a separable subset. Given positive finite measures μ, ν with equal mass and finite first moment, concentrate both on a closed separable subspace $S \subset M$. Passing to the completion $\widehat{S}$, the measures become measures on a Polish metric space. The source Theorem 1.13 gives finite-support approximants in the Kantorovich–Rubinstein norm on $\widehat{S}$. Any finitely many atoms lying in $\widehat{S} \setminus S$ can be moved to nearby points of S , changing the transport cost by an arbitrarily small amount. Therefore $\text{TP}(M)$ is dense in $(\mathcal{M}, \|\cdot\|_{\text{KR}})$.

For the negative statement, Raad’s Corollary 3.7 gives, whenever the weight of M carries a nontrivial measure, a Borel measure μ with finite first moment such that $L_\mu \notin \mathcal{F}(M)$. Since functions in $\text{Lip}_0(M)$ vanish at the base point, $L_\mu = L_{\mu - \mu(M)\delta_0}$. This signed measure has total mass zero and finite first moment, hence is a KR measure element. If it were a KR-norm limit of elements of $\text{TP}(M)$, then it would also be a limit in the weaker Lipschitz-dual norm, forcing $L_\mu \in \mathcal{F}(M)$, a contradiction.

Dichotomy

Let κ_{rvm} be the least real-valued measurable cardinal, if it exists. Combining the source theorem with Raad’s results gives:

$\text{TP}(M)$ is dense for all finite-first-moment Borel measures on M

whenever $w(M) < \kappa_{\text{rvm}}$, and in particular for every metric space M if no real-valued measurable cardinal exists. Conversely, if a real-valued measurable cardinal exists, then the unrestricted density statement fails for metric spaces of weight at least κ_{rvm} .

Thus the source’s general-metric question has a sharp set-theoretic answer: below the real-valued-measurable threshold the analogue is true, while above it there are non-tight Borel measures giving counterexamples.

Scope Notes

The supporting paper does not explicitly claim to answer Remark 1.14 of arXiv:1902.10334; this packet records the reformulation. The earlier packet on tight-measure TC/KR density is preserved in `history/1902.10334_tight_measure_tc_kr_density_partial_2026.06.16/` as the elementary subcase.

References

- [1] S. Ostrovska and M. I. Ostrovskii, *Generalized transportation cost spaces*, arXiv:1902.10334, 2019.

- [2] L. M. Raad, *A Characterization of Borel Measures which Induce Lipschitz-Free Space Elements*, arXiv:2412.13319, version 4, 2025.